

Conclusions: While the majority of fractions of SBRT have IFVs <2 mm, up to 30% of fractions could not remain within the expected accuracy (<2 mm). Differences in IFV were seen according to immobilization device, with the Body Frame producing the most favorable results. Overall, 5 mm target margin is appropriate compensation for IFV of TB variation using current image guidance and patient immobilization methods. Additional reduction of this margin will require more dedicated online image guidance techniques.

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247 Rapid Delivery of Stereotactic Radiotherapy for Peripheral Lung Tumors using Volumetric Intensity Modulated Arcs

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Purpose/Objective(s): The delivery of high dose conventional stereotactic body radiotherapy (SBRT) for peripheral lung tumors can take >20 minutes per fraction, and optimal dose distributions are difficult to achieve for nonspherical tumors. Faster treatment delivery is more patient-friendly, and can reduce intrafraction shifts in tumor position. We evaluated the use of RapidArc (RA), a novel planning and delivery technique for Volumetric Intensity Modulated Arcs, in SBRT.

Materials/Methods: RA (version 8.2.23) was used for planning and delivery of SBRT in 17 patients with Stage I lung tumors. Three risk-adapted fractionation schemes were used, prescribed on the 80% isodose, 3×18 Gy, 5×11 Gy, or 8×7.5 Gy, dependent on size of the PTV and proximity of organs at risk (OAR). The PTV sizes ranged from 6.2–275 cm³. Clinical delivery was performed using multiple arcs, as the present RA version allows maximum 999 MU per arc. Setup was performed using a tumor match on CBCT, which was repeated after 2 arcs. For dosimetric verification, patient-specific QA was performed by comparing GafChromic EBT film dose measurements in the Quasar phantom with calculations, the Quasar phantom moving in a sinusoidal pattern with the observed tumor proximity (if >5 mm). For purpose of comparison, conventional SBRT plans using 10 noncoplanar static fields were also generated for all patients, except those receiving 8×7.5 Gy. Conformity indices (CI_{80, 60}, the volume encompassed by the 80 or 60% isodose divided by PTV-volume receiving the same dose), volume of chest wall receiving more than 45 Gy (V_{45,chest}) and typical radiation delivering times were compared.

Results: In comparison to conventional plans, RA plans achieved superior dose conformity around the PTV and reduced chest wall doses. Average CI₈₀ and CI₆₀ were 1.12 and 2.17 for RA vs. 1.24 and 2.42 for conventional SBRT, respectively. The average V_{45,chest} was 1.4 cm³ for RA and 2.0 cm³ for conventional SBRT. Film dosimetry revealed that RA plans were not compromised by possible interplay between moving leaves and a moving PTV, even with a tumor motion amplitude of up to 25 mm. Typical RA delivery times (from first beam on until last beam off, excluding time for extra setup) ranged from 4.5 minutes (7.5 Gy) to 11 minutes (18 Gy). Typical conventional SBRT delivery times range between 10 and 15 minutes. In the near future, the combination of a new clinical release of RapidArc (8.6.10) which allows delivery of >999 MU per arc, and a linac allowing dose rates of 1,000 MU/minute will allow 18 Gy to be delivered in less than 6 minutes.

Conclusions: An RA allows for rapid and highly conformal delivery of SBRT for Stage I lung tumors. The shorter treatment time reduces risk of intrafraction patient motion.

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248 Outcomes with Current Technique of Proton Beam Radiation Therapy for Uveal Melanoma: The UCSF/Tumori Foundation Experience

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Purpose/Objective(s): To evaluate control and complication rates from our current standard proton beam radiation therapy (PBRT) technique of 56 GyE in 4 fractions and anterior eye structure sparing in patients with uveal melanoma (UM).

Materials/Methods: Between 1994 and 2006, 622 patients with UM were given PBRT at the University of California San Francisco/Tumori Foundation. All patients were operated upon by a single surgeon (D.C.). Forty-seven patients who underwent eye wall resection prior to PBRT will be reported separately. The remaining 575 patient charts were reviewed. During 1994–1995, 42 patients were treated with 48 GyE in 4 fractions (fx). Thereafter, 533 patients were treated with our revised dose of 56 GyE in 4 fx. In addition, treatment practice was modified to reduce ciliary body/lens dose to decrease neovascular glaucoma (NVG) rates (Daftari *et al.*, *IJ-ROBP* 1997;39:997-1010; Rajendran *et al.*, *IJROBP* 2004;60:S311-S312). Tumor location was ciliary body in 11%, 27% macular, and 62% choroidal. Median age at dx was 60 years (range, 13–94 years). Mean largest tumor diameter was 10.8 mm (2–23.7 mm). Mean tumor height was 5.1 mm (1.3–18.3 mm); 6.3% were Stage T1, 82% T2, and 11.7% T3. Median follow-up after PBRT was 41 months (range, 0.1–159 months). For patients who received 56 GyE median follow-up was 40.8 months (0.1–152 months); for the 48 GyE group, median follow-up was 72 months (8.6–159 months). Kaplan-Meier methods, chi-square test, and Cox proportional hazards model were utilized to calculate recurrence and survival rates and test for variables correlating with control and survival.

Results: Local control (LC) at 2 years was $98.1 \pm 0.6\%$ and 5 years $95.6 \pm 1.3\%$ for patients undergoing treatment with current clinical standard of 56 GyE. In the 48 GyE group, 8 of 42 patients (19%) failed vs. 20 of 533 patients (3.8%) in the 56 GyE group ($p < 0.01$). A detailed analysis was completed for our 56 GyE group. Median time to local failure was 25.5 months. Distant metastasis (DM) free survival at 5-years was $86.2 \pm 2.3\%$. Median time to DM was 38.6 months (4.8–92 months). Overall survival at 5-years was $84.8 \pm 2.3\%$. None of the patients with T1 tumors recurred during follow-up. The enucleation rate was 7.5% and the NVG rate was 6.6%. On multivariate analysis, macular location ($p = 0.014$) and ciliary body involvement (CBI) ($p = 0.013$) were poor prognostic features for LC. T-Stage ($p = 0.07$) and CBI ($p = 0.001$) were poor prognostic features for OS.

Conclusions: Excellent local control rates, 98% at 2 years and 96% at 5 years, have been maintained with proton therapy for patients with uveal melanoma. The lower dose of 48 GyE resulted in lesser control. Our current revised clinical practice of PBRT for